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Final Paper | Phil of AI, Callard & Bridges

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Limits of Matter and Manner

The emergence of “mechanistic interpretability”, the study of how neural networks

compute their outputs, rests on the idea that, at the moment, the “black box”

description of generative AI systems isn’t sufficient. Essentially, however complex

artificial intelligence models are, reverse engineering their outputs can reduce the

opacity of generative machine learning models and show a mechanical, identifiable

process of understanding or thinking that takes unseen inputs and transforms them

into intelligible, and naturalistic-seeming outputs.

NOT CLEAR ABOUT ALL THIS: ISN’T THE OPACITY IN QUESTION A MERE OPACITY ABOUT THE DETAILS OF THE OPERATIONS OF THE MECHANICAL PROCESS, AND NOT OPACITY ABOUT THE FACT THAT IT IS A MECHANICAL PROCESS? BUT MAYBE YOU AREN’T SAYING OTHERWISE...

GPT models have been passing the Turing Test, and we speak about AI as if it is seeing, painting, drawing, thinking, learning, and communicating; our communication with chatbots has even led us to see

the production of these outputs as a type of companionship, arguably an anthropomorphism of artificial intelligence. Still, more than three-quarters of US adults

are either unsure of the benefit or sure that AI may be harmful. So, why do we tend to

anthropomorphize and rationalize AI systems? Is this tendency justifiable, or is it a

blinding human tendency that traps us into believing that humans and AI are

intellectually matched?

In this paper, I will argue that the architectural complexity (the construction) of

Generative AI forces us to understand these systems as intentional out of pragmatic

necessity – to treat them as unintentional would be to call their outputs irrational or

untrustworthy.

OK INTERESTING

But, this pragmatic necessity masks the failure of Generative AI to

satisfy a construction-level characterization required for understanding, intentionality,

and rationality. The tension between the complexity and opacity of AI models and the

pragmatic necessity we have to understand and engage with them is what traps us into

the illusory anthropomorphization of AI as rational and intentional interlocutors.

SO JUST TO BE SURE: YOU'RE SAYING WE CAN'T HELP THINKING SOMETHING WE KNOW IS FALSE? (THIS IS A PARADOX—BUT TO SAY THIS IS NOT TO SAY IT ISN'T THE SITUATION WE FIND OURSELVES IN!)

In his essay, "Intentional Systems", Daniel Dennett defines Intentionality as any system

"whose behavior can be (at least sometimes) explained and predicted by relying on

ascriptions to the system of beliefs and desires".

I THINK DENNETT'S DEFINITION IS INCOHERENT. IT'S LIKE SAYING: "A DOG IS ANYTHING THAT PEOPLE USEFULLY TAKE TO BE A DOG." WE CAN ASK: TAKE TO BE A WHAT? THE WORD DOG SHOWS UP ON BOTH SIDES OF THIS ANALYSIS. SAME FOR DENNETT'S DEFINITION, IT SEEMS TO ME.

He claims that intentionality is

"observer dependent"- that intentionality is defined only within that perspective of

the inquiring party. He also assumes intentionality as a "feature of linguistic entities"

,

which allows me to include Large Language Models in the discussion of intentionality.

And, finally, Dennett assumes that the system in question is behaving rationally given

its goals. Essentially, to seriously inquire about a system's Intentionality, until you end

your investigation, you should assume the system's rationality says something about

how to predict its behavior. I wonder if this is because Dennett contends that rationality

is something we either have not yet been able to define or is less philosophically

problematic than Intentionality. In other words, Intentionality is some higher barrier to

the human condition that a system must surpass to be comparable to the humankind.

Dennett also situates the Intentional Stance in opposition to the Physical and Design

Stances as strategies for classifying and explaining a system's behavior. The proposed

distinction between these stances is the mechanism of prediction. Prediction is

enabled by the laws of nature by the Physical Stance, the functional design of the

system by the Design Stance, and the assumption of rationality by Dennett's Intentional Stance. Thus, only a rational system, where rationality is explicitly not

definable through the Physical or Design Stances, can be intentional. So, under

Dennett's definitions, to determine if AI is Intentional, we must first believe that an AI's

output should be assessed on the basis of rationality.

Dennett does not undertake defining why doing the optimal thing given a specific goal

is a sufficient description of rationality in this essay, a smart move considering the

difficulty of that inquiry for mankind, the rational kind. The behavior of the system is

thus seen as something too complex, or too massive of a concept for the Physical and

Design Stances to define: brains and computers are both systems in which there are too

many physical variables for us to keep track of, thus failing to be explained by the

Physical Stance, and in which a brain or computer's internal states have too intricate of a representation to be described by the Design Stance. Whether or not the Physical or

Design Stance is intelligible is unimportant because they both fail to be pragmatic

stances. Humans don't have the computational power to predict a complex system's

behavior physically – this would be “a pointless and herculean labor” according to

Dennett. Similarly, the internal operations of the human biological system or Large

Language Model are so layered and multimodally impacted that predicting behavior

becomes cognitively impossible through Design Stance inspection.

Importantly, these failures of the Design and Physical Stances are unfortunate to

Dennett because they are greatly pragmatic stances for less complex systems; they are

only limited by our own computational and biological limitations. But, it is exactly

those limitations that Dennett believes make the Intentional Stance the pragmatic, or at

least justifiable, approach to studying behavior. How can we claim to have a theory of

behavioral prediction that our own behavior (our limitations) prevents us from using?

Dennett believes we should have no such theory. Thus, the value of the Intentional

Stance is an acceptance of our limited capacities, using our ability to exhibit rationality

through our behavior to produce a reasonable theory of behavioral prediction.

GOOD

To examine a system through the Intentional Stance is to only use our human-granted

rationality without having to discuss its substance or the existence of consciousness.

Dennett makes an arguably compelling argument for the viability of the Intentional

Stance and, for the reasons I discussed above, proves the Intentional Stance to be a

pragmatic necessity for understanding complex systems. Notably, the Intentional

Stance does not metaphysically grant AI a soul or a human likeness, but offers a

strategy to evaluate predictive behavior.

THOUGH AS I MENTIONED EARLIER, ONE MIGHT THINK THAT THE CONTENT OF THE STRATEGY IS INCOHERENTLY CIRCULAR.

The Intentional Stance seems to be quite

similar to how we already evaluate AI. We don't say "ChatGPT is acting as if it

understood my question" when it gives us satisfying answers; we say "ChatGPT

understood my question"

. Thus, the Intentional Stance seems to be a natural mode of

investigation for us. This is good for us, considering the Physical and Design Stances' practical difficulties. But, is the pragmatic utility or practical success of the Intentional

Stance enough to let it say something about the behavior of understanding?

Jason Bridges argues that genuine understanding requires an "activity-level" and

"construction-level" characterization of the system in his paper,

"Meaning &

Understanding"

. This description responds to an intuitive definition of understanding:

the Guidance Conception says that understanding is the process of having something

"come before the mind" to guide one's actions. The major flaw in the Guidance

Conception, for Bridges, is that the directions of a guide must be understood by the

system following them. So, we cannot use a guide to define understanding if understanding is part of the process needed to follow a guide. Ironically, this seems

similar to the structure of Dennett's defense of the intentional Stance: we cannot use

the Physical or Design stance to define understanding if we are unable to understand

how to pragmatically use these stances. form of correctness, helpfulness, or precision.

This characterization is context-dependent, and though necessary, only describes

"accordance"

, or what many of my computer scientist friends would call "spurious correlations"

. Thus, the construction-level characterization provides a context-independent description of the understanding system in question. The description of “apparatus” or architecture of an internal state of a system (construction-level) must correspond to the observed understanding at the activity-level under Bridges' requirements for an understanding Mechanism. When a Large Language Model (LLM) produces an output, we ought to say, by the Intentional Stance, that the LLM decided, or reasoned, to answer our question. In other words, the LLM was guided towards an answer by its own rationality. For Bridges, this is an activity-level characterization. By Dennett’s design, the activity-level characterization contains a projection of our own reality onto a Mechanism’s outputs, but Bridges argues a construction-level characterization must be provided. The field of mechanistic interpretability seems to provide this: examinations of the weights, biases, and layered architecture of the neural networks that underlie LLMs. But this construction-level characterization is unsatisfactory because it arguably remains context-dependent (although that is not the goal of mechanistic interpretability researchers). If mechanistic interpretability is only able to use the context-rich activity-level characterizations to support construction-level characterization, then we do not isolate the apparatus of a model from its context. If this field of study continues



on to successfully find inherent conceptual meaning in the states of AI models,

generalizable across different machine learning architectures, then mechanistic

interpretability will be the true study of computational understanding.

To make this conflict tangible, consider the activity-level characterization of an LLM's

output: that it "gave a reasonable or helpful response" to a question. The supposed

construction-level accounts are the states of a neural network: matrices of numbers

and coefficients (weights) of variables. There is no context-independent mapping of

"understanding" or "helpfulness" to a matrix of numbers. Recall that Dennett defined

Intentionality as a feature of linguistic entities, whereas the LLM's actual mechanism is

purely numerical. Thus, the LLM satisfies the conditions for Intentionality as a

pragmatic simulation. The construction level remains mathematically/numerically

abstract rather than conceptually grounded, so the system is fully secluded to

activity-level characterizations. The disconnect here is that the Mechanism produces

the outputs of understanding without satisfying the conditions for a complete process

of understanding required by Bridges.

I am unsure of whether or not humans pass Bridge's test, but it seems silly to say that

we are projecting a higher form of understanding to understand our own...

understanding. Still, this seems to be the project of the biological sciences and

scientific inquiry in general. In this essay, I have tried to argue why AI does not satisfy

Bridges requirements for understanding. So far, we have yet to reach satisfying

construction-level characterizations of AI systems, leaving us with an activity-level

characterization that projects rationality onto these systems — the aforementioned

anthropomorphization. The complexity of AI systems, like ChatGPT, forces us into

taking the Intentional stance.

THOUGH THE “STANCE” IS NOT A JUDGMENT ABOUT THE SYSTEM, RIGHT?

But, the opacity that comes with architectural complexity also prevents us from identifying a construction-level characterization. The more complex AI models get, the more we get backed into a corner granting AI a “pinched

and artificial” rationality. And we are thus pushed even further from crafting a

construction-level understanding. Our anthropomorphism of AI is explainable by the

limitations put on our brains and bodies by cognitive capacity, a testament towards

only humankind’s Intentional and understanding capacities.

RHEA, THIS IS A VERY GOOD, THOUGHTFUL EXPLORATION OF THE INTENTIONAL STANCE (AND OUR INABILITY TO AVOID IT, IN THE CASE OF LLM'S), AND HOW THAT STANCE RELATES TO MEANING/UNDERSTANDING. REALLY NICE WORK. GRADE: A-